



RAJARATA UNIVERSITY OF SRI LANKA

Faculty of Technology-Department of ICT

Bachelor of Information and Communication Technology (BICT)

Module: ICT 1209 - Web Technologies

MINI PROJECT: Final Report

Project Title: Xnotes - The BICT Resource Hub

Theme: Student Resource Hub

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Date: 26th September 2026

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1. Project Overview

The primary objective of "**Xnotes**" is to develop a centralized, interactive web application that facilitates the efficient sharing, organization, and retrieval of academic resources.

Target Audience: Bachelor of Information Communication Technology (BICT) Undergraduates at the Faculty of Technology Rajarata University of Sri Lanka

Key Features:

- Secure User Authentication (Login, Signup with OTP email verification).
- Password Recovery via OTP.
- Dashboard for users to manage and delete their uploaded notes.
- Secure File Upload system with strict MIME-type and size validation (Max 20MB).
- In-browser PDF Viewer for reading notes without downloading.
- Search functionality and dynamic popular categories.

2. System Design

➤ Database Schema (xnotes)

The database consists of three main tables:

Table Name	Description	Fields
users	Stores user accounts	id, username, email, password, created_at
notes	Stores file metadata	id, user_id (Foreign Key), module_code, title, category, file_path, uploaded_at
messages	Stores contact form submissions	id, name, email, message, created_at

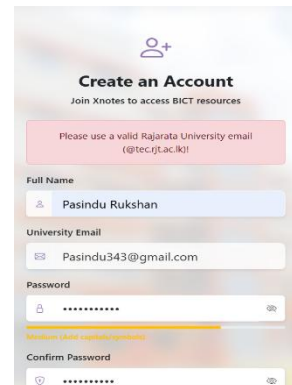
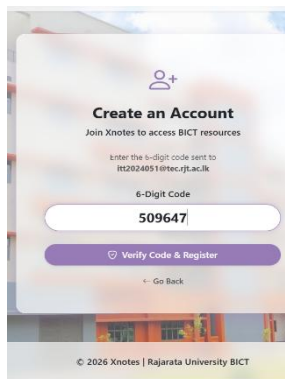
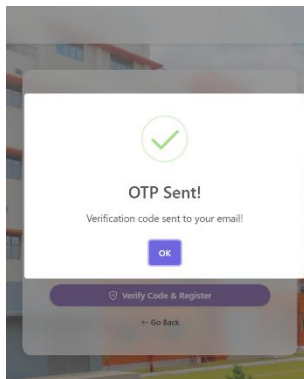
➤ Folder Structure

- /auth: Contains authentication scripts (login.php, register.php, forgot-password.php, logout.php).
- /css: Stylesheets including Glassmorphism effects (style.css).
- /images: Logos and background assets.
- /includes: Backend dependencies (db.php, PHPMailer/).
- /js: JavaScript functionality (script.js).
- /uploads: Directory where user files are securely stored.
- Root: Main pages (index.php, dashboard.php, upload.php, contact.php, view-notes.php, view-file.php).

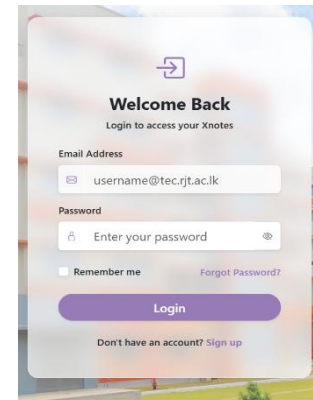
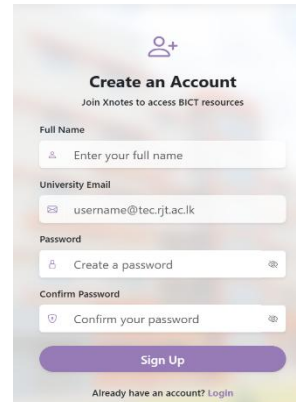
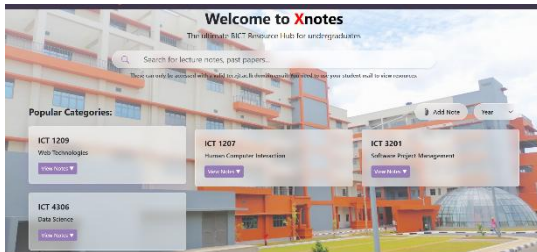
4. Implementation

(Note: Add screenshots for each of these features before submitting the PDF)

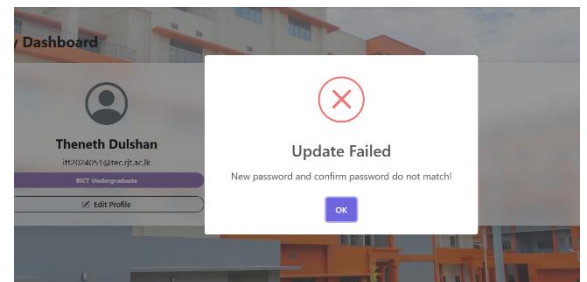
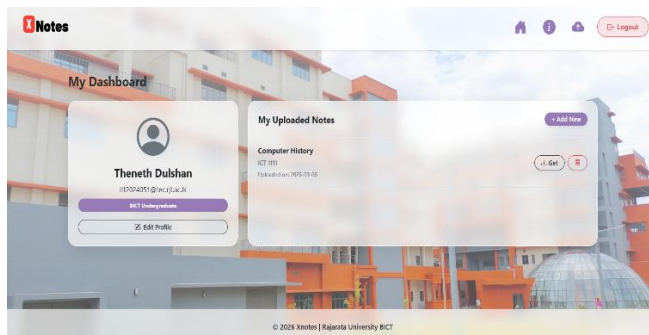
- Authentication & OTP (auth/register.php, auth/forgot-password.php): Implemented using PHP sessions and PHPMailer. Restricts registration to university emails and verifies ownership using a 6-digit OTP.



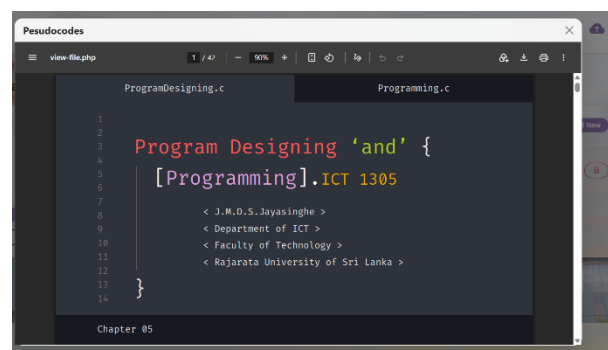
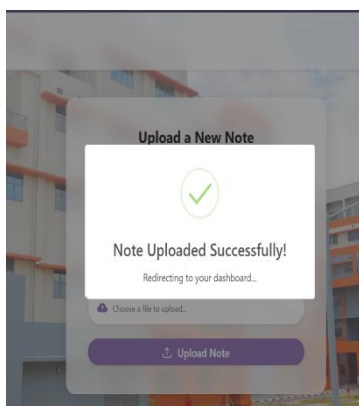
- Glassmorphism UI (css/style.css): The entire application utilizes a modern, frosted-glass design pattern using CSS backdrop-filter, providing a clean and engaging user experience.



- File and Profile (upload.php, dashboard.php): Users can upload notes. The backend uses finfo_file to verify the actual MIME type of the uploaded file to prevent .exe or malicious file uploads disguised as PDFs. Also, User can manage and edit user's profile.



- In-Page PDF Viewer (view-notes.php): A custom modal using a Bootstrap modal and an iframe allows students to read PDF notes directly in the browser without cluttering their device storage.



5. Challenges and Solutions

Challenge	Solution
Preventing outsiders from creating accounts.	Implemented a strict domain check (<code>str_ends_with</code>) forcing users to register with a <code>@tec.rjt.ac.lk</code> email, combined with SMTP OTP verification to ensure they actually own the email.
Security risks with file uploads (e.g., users renaming viruses to <code>.pdf</code>).	Instead of just checking the file extension, we used PHP's <code>finfo_open(FILEINFO_MIME_TYPE)</code> to inspect the file's raw bytes and verify it is a genuine document, alongside a 20MB size limit constraint.
GitHub merge conflicts when collaborating.	Divided our tasks logically (one handling UI/modals, the other handling DB/backend), and used Git branching and careful conflict resolution to merge code.

6. Individual Contribution

- **W.P.T.D. Jayawardhana (theneth-ai):**
 - Designed the core database schema and configured the secure PDO MySQL database connection.
 - Developed the interactive Dashboard for note management, enabling users to view and securely delete their uploaded files.
 - Implemented the secure file upload logic with MIME type validation and file size constraints and secure file remaining.
 - Built the search algorithm and filtering system (Year/Category) for the homepage using JavaScript and PHP.
 - Handled the overarching backend architecture and advanced session management (including 'Remember Me' auto login feature), and GitHub repository management.

- Integrated SweetAlert2 to create the dynamic file upload loading indicator, and designed the fundamental structured framework for the UI.
- Design the basic Css Glassomorphism UI
- **H.P. Rukshan (hprukshan):**
 - Styled and refined the complete Glassmorphism UI across all pages, integrating advanced CSS effect and ensuring Bootstrap 5 responsiveness.
 - Implemented the secure OTP email verification system using PHPMailer for secure User Registration and Password.
 - Created the interactive in-page PDF Modal viewer for seamless, on-the-fly note reading without leveling the dashboard.
 - Developed real-time frontend form validation (JavaScript) and the interactive password strength indicators.
 - Built the Contact Us form integration and enhanced the dynamic navigation bar states.
 - Configured general SweetAlert2 custom animations and modern alert popups for user interactions (Success/Error messages).

Note: We collaboratively developed the overall web application, engaging in regular pair-programming sessions to debug, integrate the backend-frontend components, and optimized the project features by supporting each other.

7. Setup Instructions

To run this project locally using XAMPP:

1. Copy the entire project folder into C:\xampp\htdocs\.
2. Open XAMPP Control Panel and start Apache and MySQL.
3. Open a browser and go to <http://localhost/phpmyadmin>.
4. Create a new database named xnotes.
5. Click Import, select the database.sql file provided in the root directory, and execute to create the tables.
6. Open auth/register.php and auth/forgot-password.php and configure the SMTP_PASS with a valid Gmail App Password to test email functionalities.
7. Ensure extension=openssl is uncommented in your XAMPP php.ini file.
8. Access the project at: <http://localhost/project/index.php>.
9. Github link: <https://github.com/theneth-ai/Student-Resource-Hub---ICT-1209-Mini-Project>

8. References

- [1] Bootstrap, “Bootstrap 5 Documentation.” [Online]. Available: <https://getbootstrap.com/docs/5.3/>. [Accessed: July. 1, 2026].

- [2] PHPMailer, “PHPMailer GitHub Repository.” [Online]. Available: <https://github.com/PHPMailer/PHPMailer>. [Accessed: Sep. 2, 2026].

- [3] PHP Group, “PHP Official Manual (PDO, Sessions, finfo).” [Online]. Available: <https://www.php.net/manual/en/>. [Accessed: Aug. 15, 2026].

- [4] SweetAlert2, “SweetAlert2 Documentation.” [Online]. Available: <https://sweetalert2.github.io/>. [Accessed: Sep. 4, 2026].